East West University

Department of Computer Science & Engineering

CSE350: Data Communication

Section: 01 | Spring - 26

Mini-Project Report

Development of a Multi-Campus Subnet-Based
University Network Architecture

Submitted By

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1. Problem Statement

Avalon University is a multi-campus educational institution that provides wired internet access across classrooms, laboratories, administrative offices, and libraries. The university runs several critical network-dependent services, including online admissions, academic advising, results management, e-Tendering, library services, and financial accounts.

The existing setup lacked a structured inter-campus network. Each campus operated in isolation, which made sharing data between departments or running university-wide services difficult. The goal of this project is to design a complete, routed network that connects all four campuses so that any host can communicate with any other host across the university.

The specific requirements are:

- Four campus LANs, each with wired hosts covering all university facilities
- Four routers, one per campus, connected using static routing
- IP addressing using Class B for LAN subnets and Class C for inter-router (WAN) links
- Verified end-to-end connectivity through successful ping operations across at least three LANs

2. Design & Experimental Results

2.1 Network Topology

The network uses a linear chain topology: Router0 (Main) connects to Router1 (North), Router1 connects to Router2 (South), and Router2 connects to Router3 (East). Each router also connects to its own campus switch and host machines. Static routes are configured on every router so that packets can travel across the full chain.

Each campus has one switch serving all local hosts. The switch connects to the campus router via a FastEthernet link. Inter-router links use Serial interfaces with clock rates set to 64000 bps on the DCE side.

2.2 IP Addressing Scheme

Class B addresses (128–131.10.0.0/16) are assigned to each campus LAN. Class C addresses (192–194.120.10.0/24) are used for the three serial WAN links between routers.

LAN Subnets:

Campus	Network	Subnet Mask	Gateway
Campus1(main) (R0)	128.10.0.0	255.255.0.0	128.10.7.16
Campus2(Second)(R1)	131.10.0.0	255.255.0.0	131.10.7.16
Campus(Third) (R2)	129.10.0.0	255.255.0.0	129.10.7.16
Campus(Fourth) (R3)	130.10.0.0	255.255.0.0	130.10.7.16

WAN Links (Inter-Router Serial):

Link	Network (Class C)	Router A IP	Router B IP
R0 ↔ R1	192.120.10.0/24	192.120.10.1	192.120.10.2
R1 ↔ R2	193.120.10.0/24	193.120.10.1	193.120.10.2
R2 ↔ R3	194.120.10.0/24	194.120.10.1	194.120.10.2

2.3 Static Routing Configuration

Each router has static routes pointing to all remote networks. The next-hop address is always the directly connected serial interface of the neighboring router. Below are the routing tables for all four routers as configured in Cisco IOS CLI.

Router0 (Main Campus) — Static Routes

Destination	Mask	Next Hop
193.120.10.0	255.255.255.0	192.120.10.2
194.120.10.0	255.255.255.0	192.120.10.2
129.10.0.0	255.255.0.0	192.120.10.2
130.10.0.0	255.255.0.0	192.120.10.2
131.10.0.0	255.255.0.0	192.120.10.2

Router1 (North Campus) — Static Routes

Destination	Mask	Next Hop
128.10.0.0	255.255.0.0	192.120.10.1
194.120.10.0	255.255.255.0	193.120.10.2
129.10.0.0	255.255.0.0	193.120.10.2
130.10.0.0	255.255.0.0	193.120.10.2
193.120.10.0	255.255.255.0	193.120.10.2

Router2 (South Campus) — Static Routes

Destination	Mask	Next Hop
192.120.10.0	255.255.255.0	193.120.10.1
128.10.0.0	255.255.0.0	193.120.10.1
131.10.0.0	255.255.0.0	193.120.10.1
130.10.0.0	255.255.0.0	194.120.10.2

Router3 (East Campus) — Static Routes

Destination	Mask	Next Hop
192.120.10.0	255.255.255.0	194.120.10.1
193.120.10.0	255.255.255.0	194.120.10.1

128.10.0.0	255.255.0.0	194.120.10.1
129.10.0.0	255.255.0.0	194.120.10.1
131.10.0.0	255.255.0.0	194.120.10.1

2.4 Router CLI Configuration Screenshots

The following screenshots show the actual IOS CLI commands used to configure each router interface and static routes inside Cisco Packet Tracer.

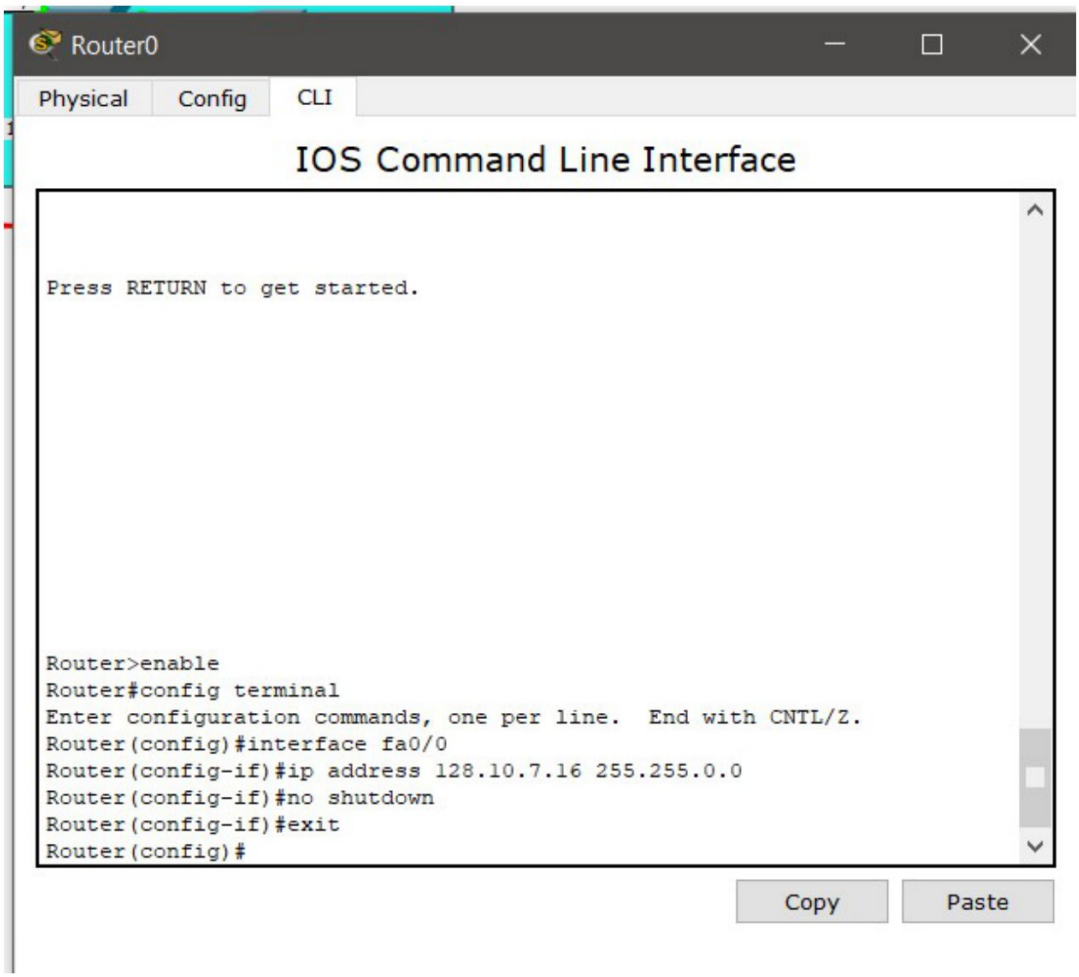

Figure 1: Router0 — FastEthernet0/0 LAN interface configuration (128.10.7.16/16)

Figure 2: Router0 — Serial2/0 WAN interface configuration (192.120.10.1)

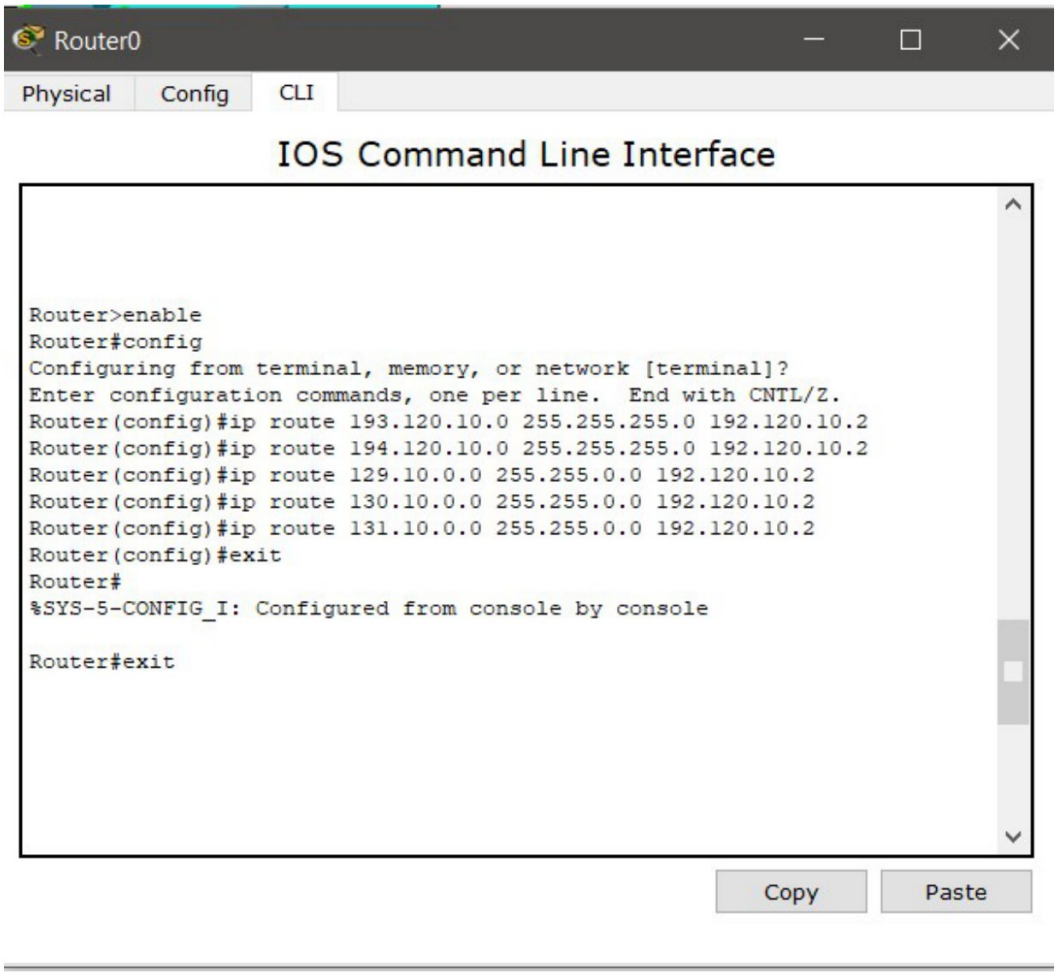

Figure 3: Router0 — Static route entries

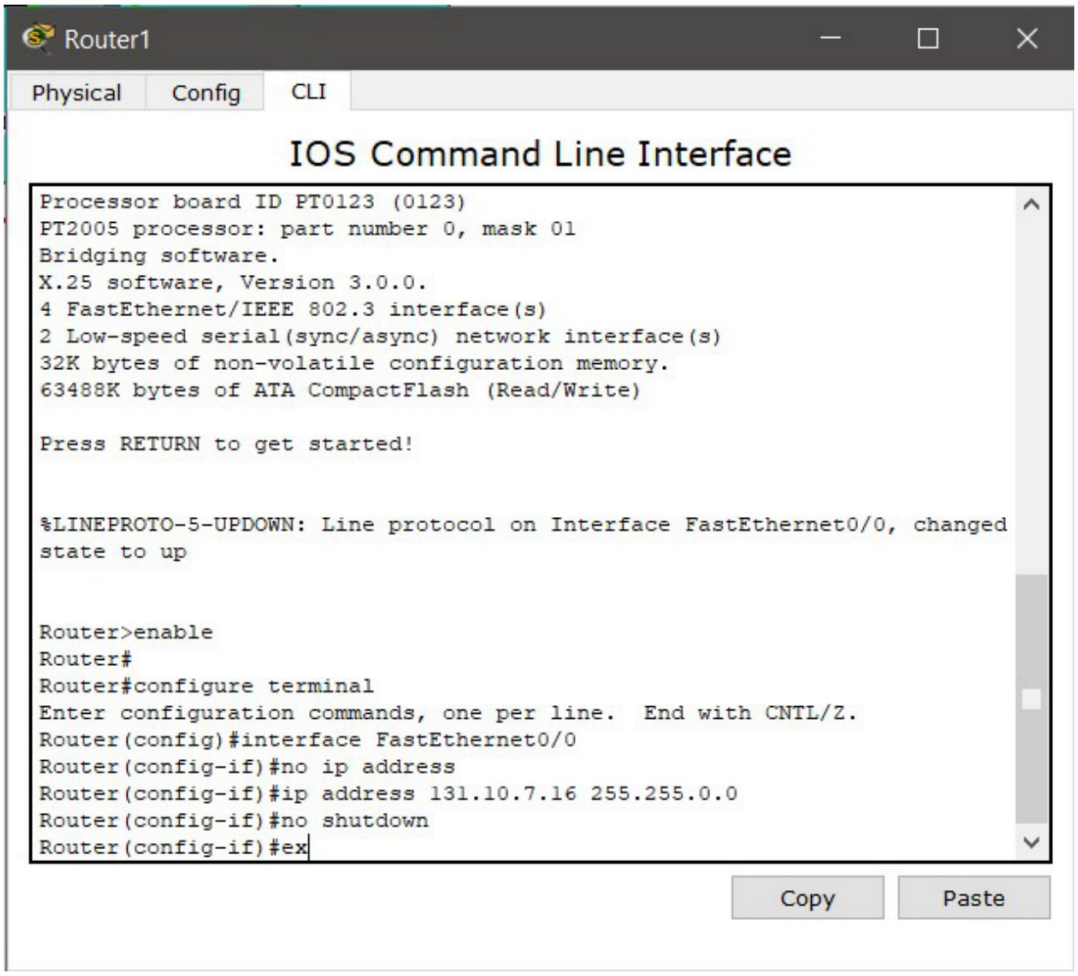

Figure 4: Router1 — FastEthernet0/0 LAN interface configuration (131.10.7.16/16)

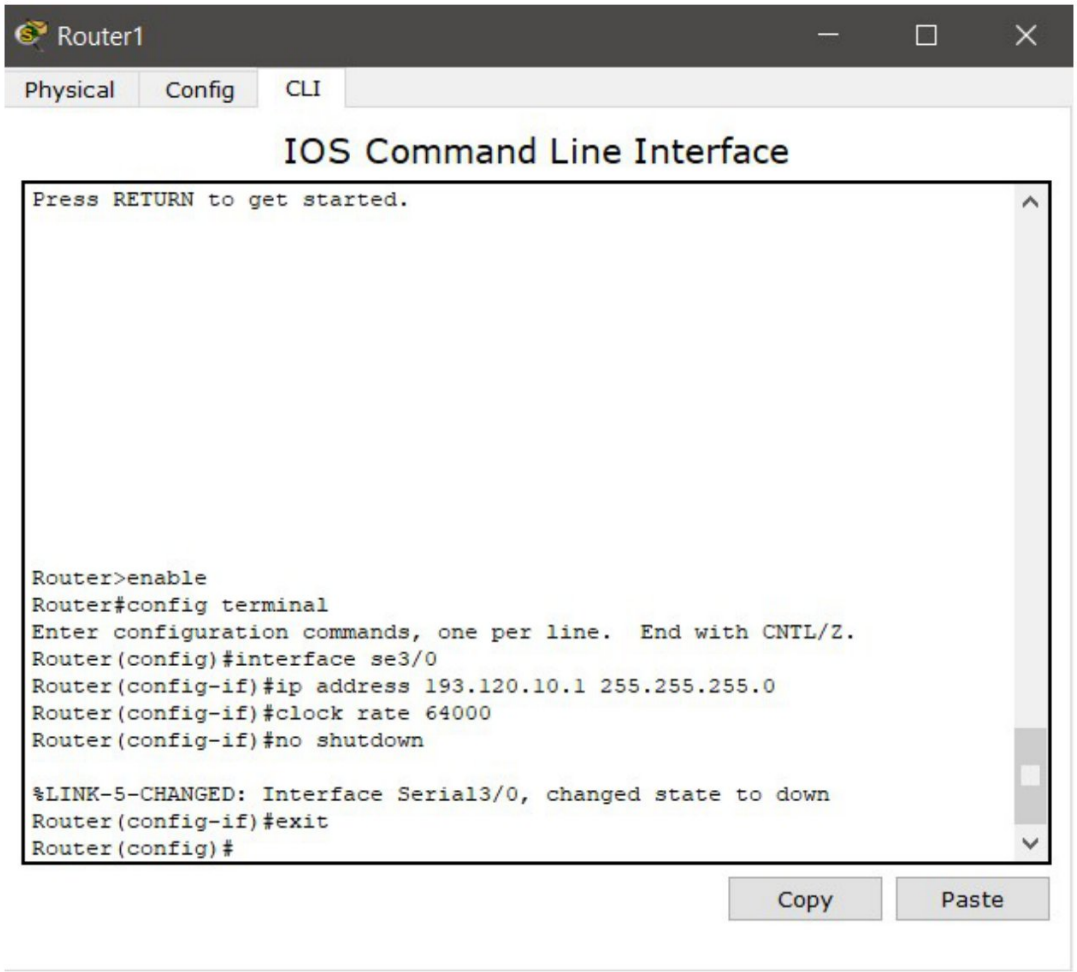

Figure 5: Router1 — Serial3/0 WAN interface configuration (193.120.10.1)

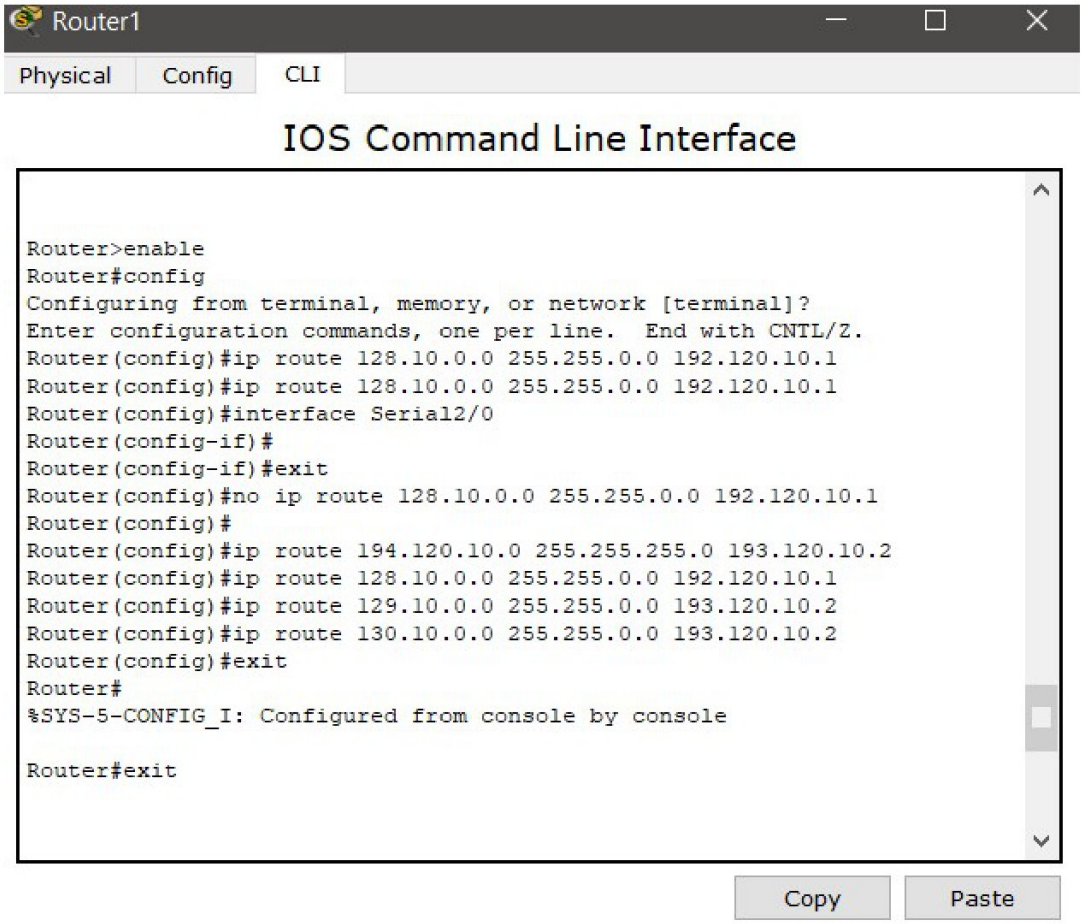

Figure 6: Router1 — Static route entries

Figure 7: Router2 — FastEthernet0/0 LAN interface configuration (129.10.7.16/16)

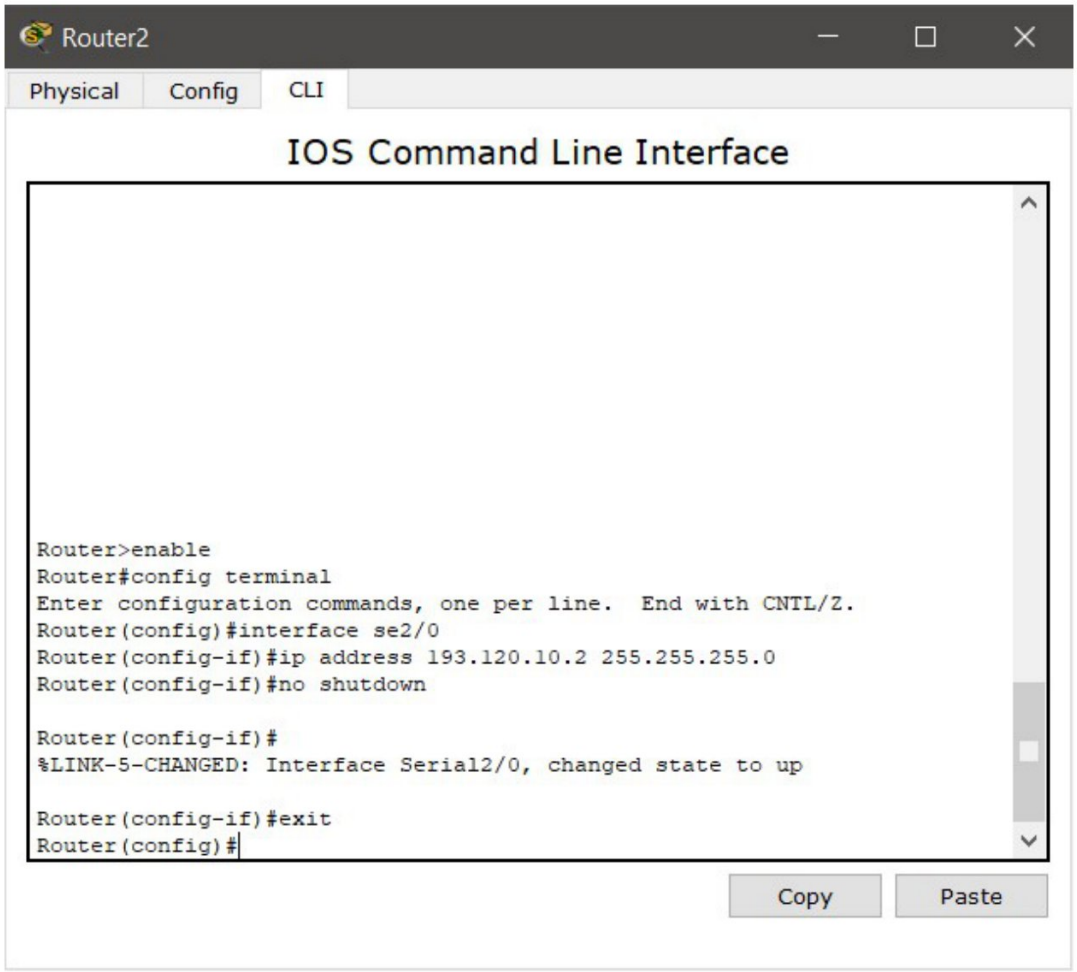

Figure 8: Router2 — Serial2/0 WAN interface (193.120.10.2) configuration

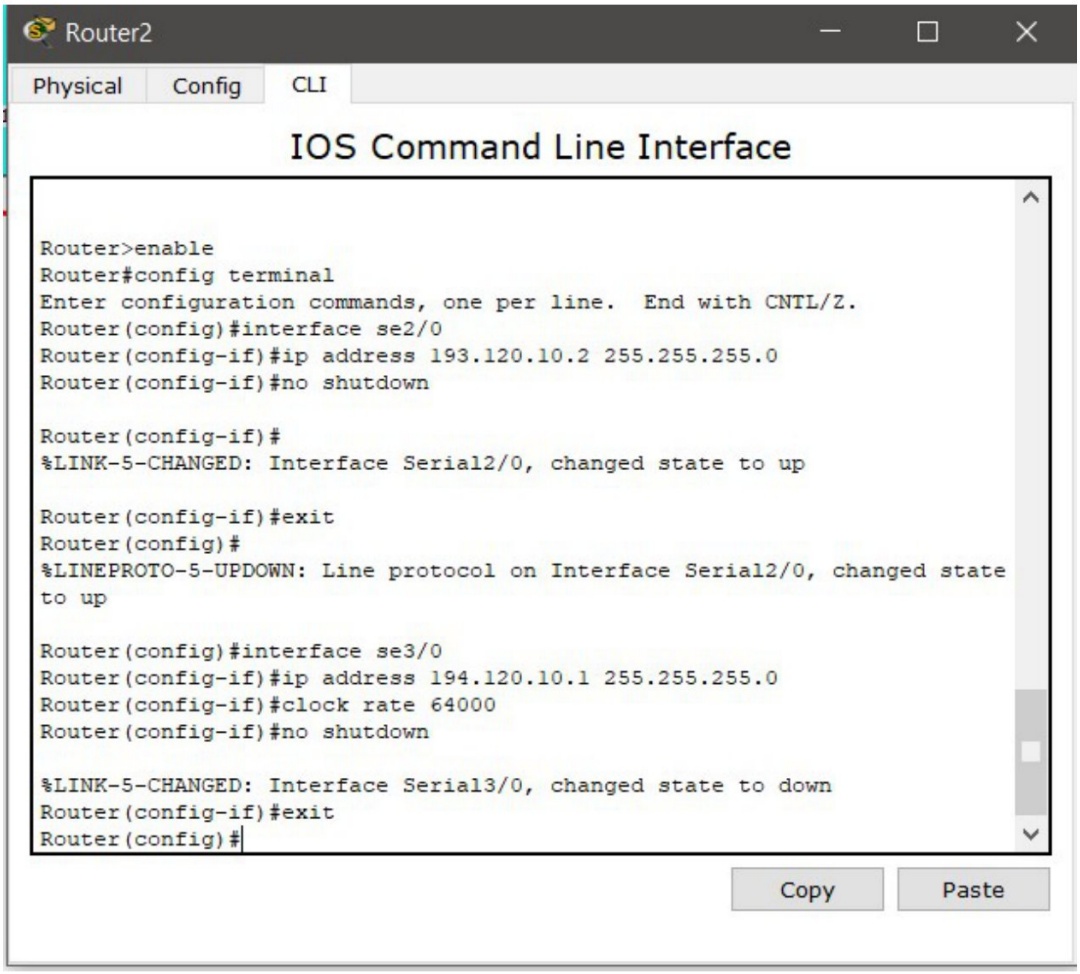

Figure 9: Router2 — Serial3/0 WAN interface (194.120.10.1) configuration

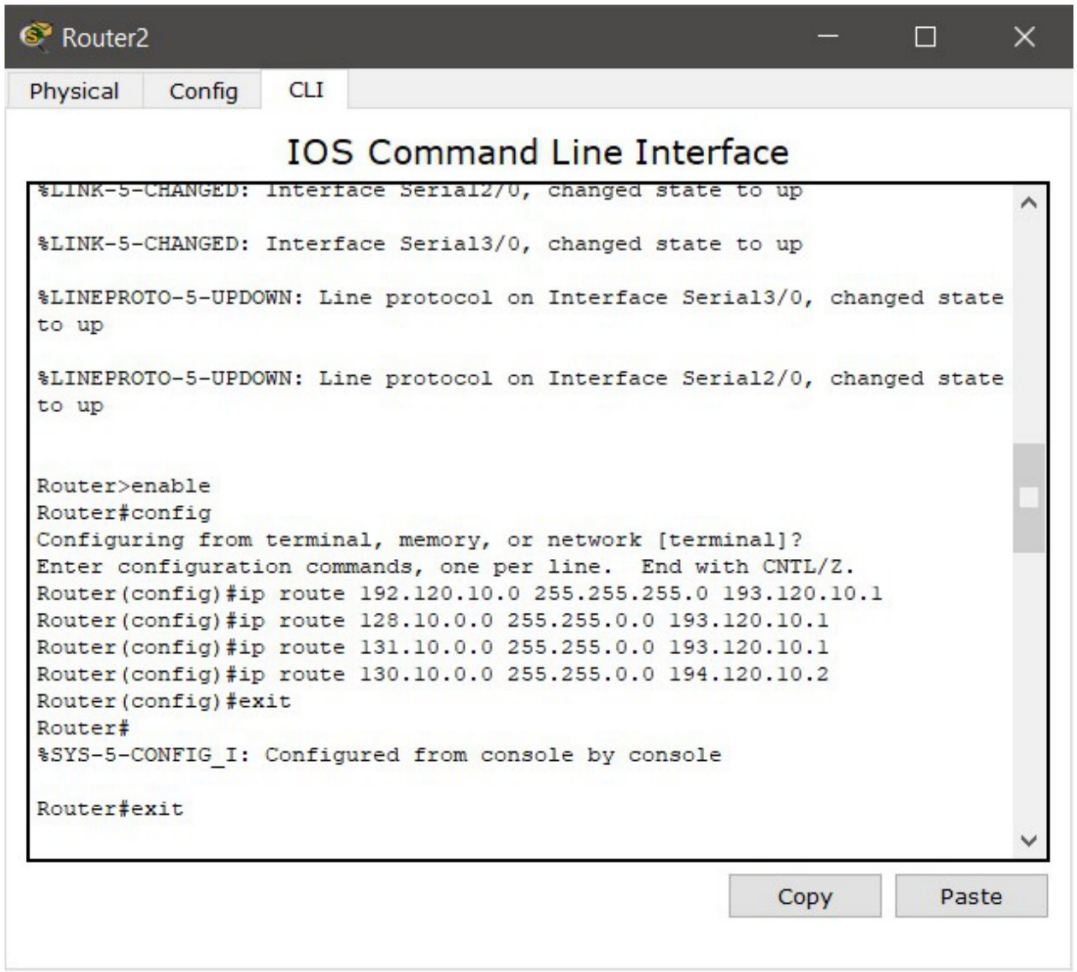

Figure 10: Router2 — Static route entries

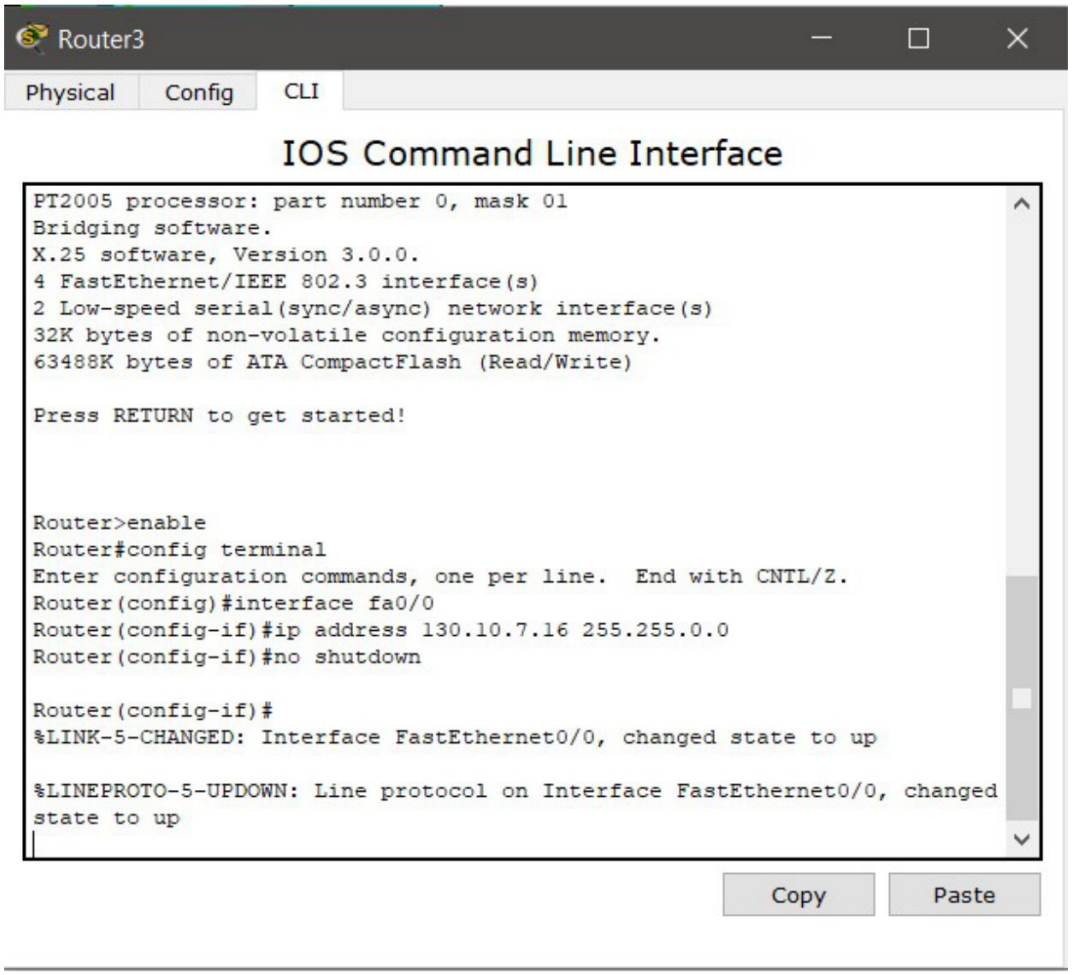

Figure 11: Router3 — FastEthernet0/0 LAN interface configuration (130.10.7.16/16)

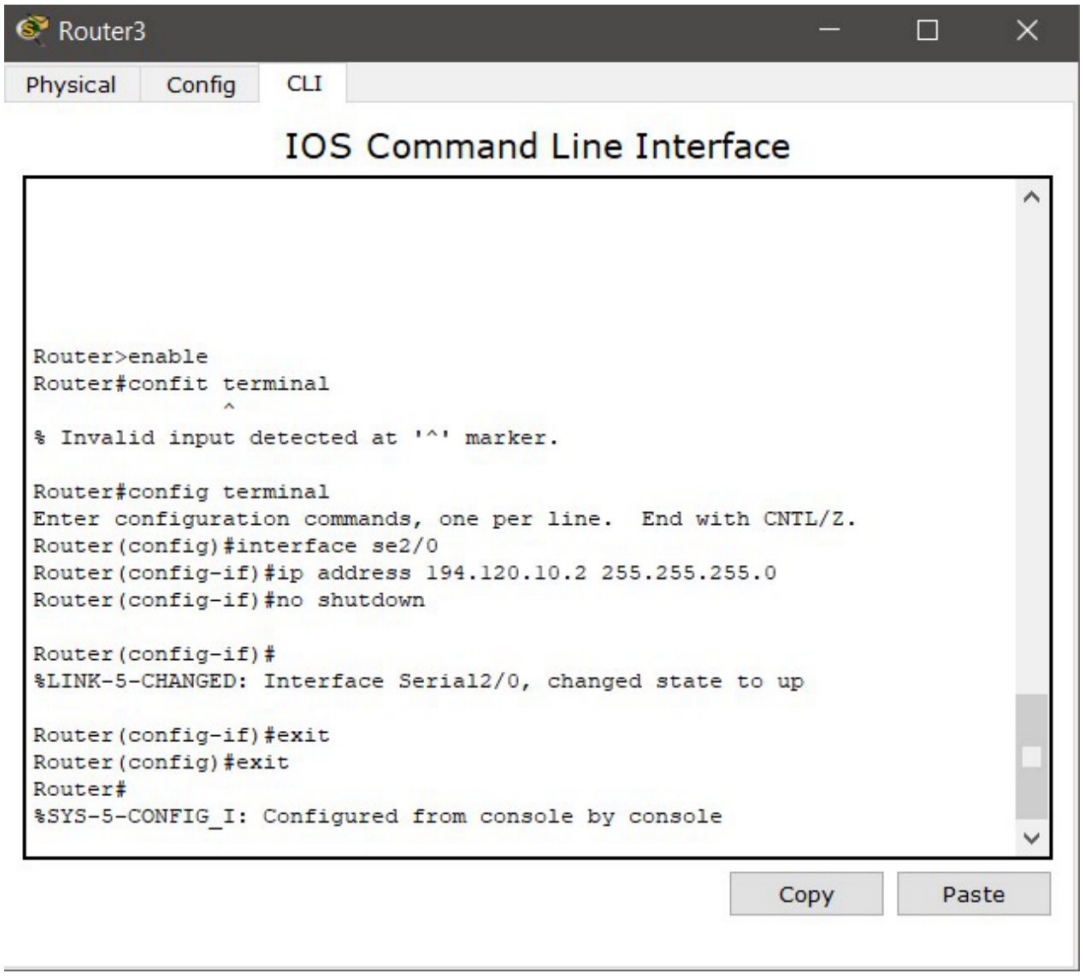

Figure 12: Router3 — Serial2/0 WAN interface configuration (194.120.10.2)

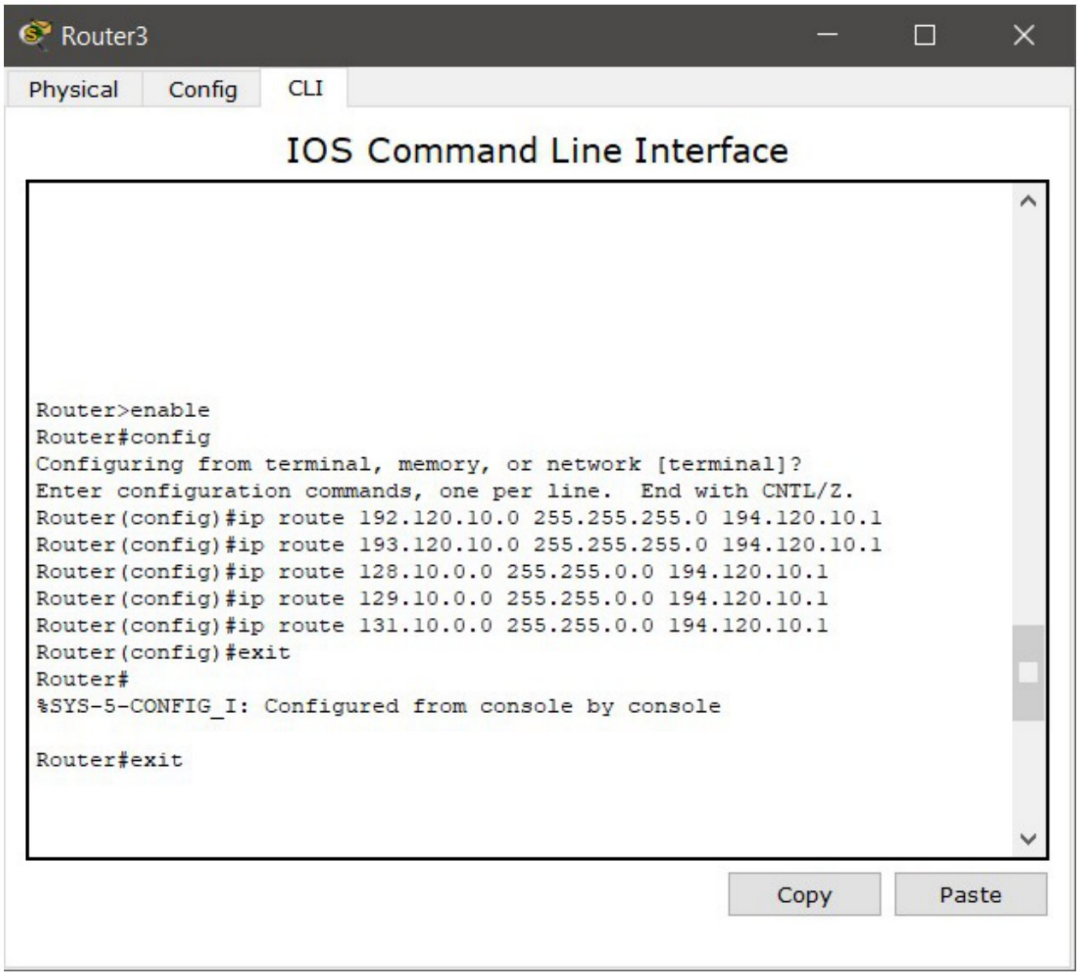

Figure 13: Router3 — Static route entries

2.5 Connectivity Verification — Ping Results

After configuring all interfaces and static routes, ping tests were run from hosts across different campuses to confirm end-to-end connectivity. A 25% packet loss on the first ping is normal in Packet Tracer because the first packet triggers ARP resolution. Subsequent packets succeed normally.

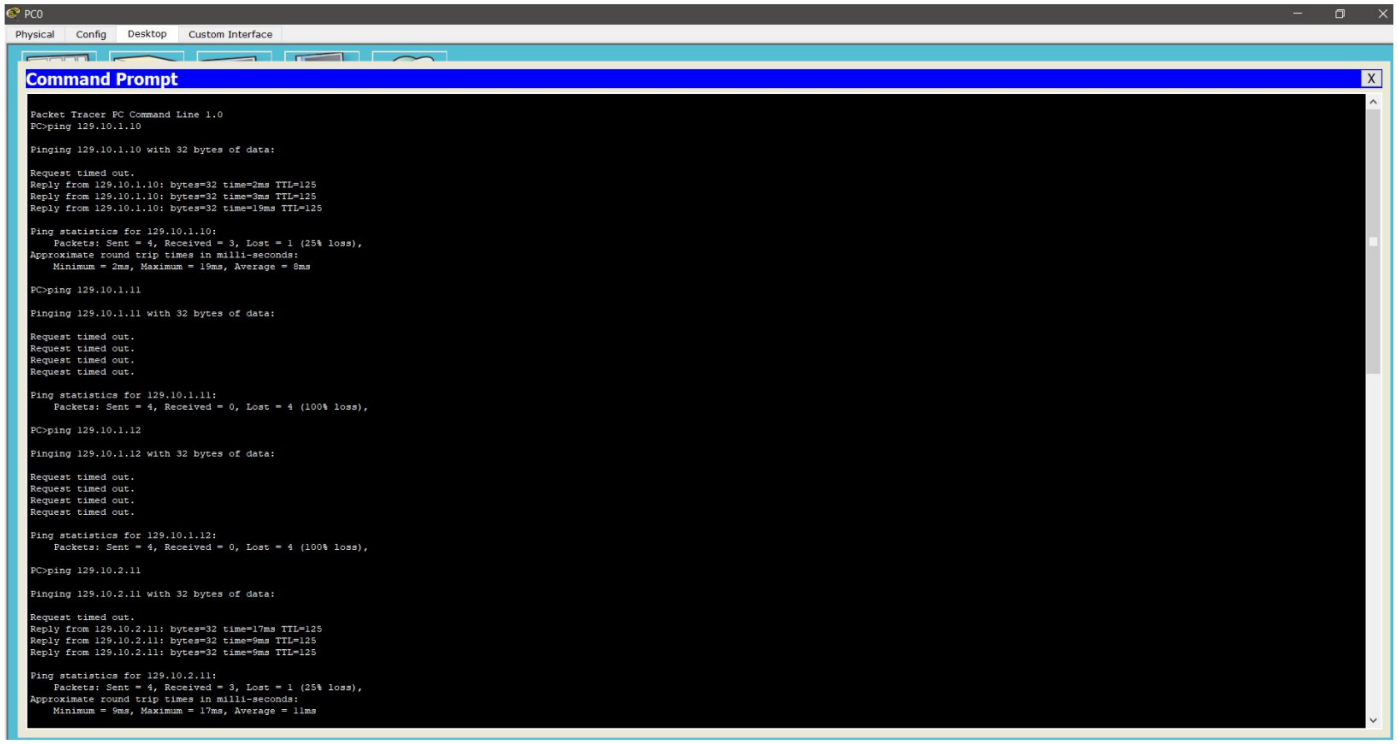

Figure 14: PC0 (Main Campus) pinging hosts on 129.10.1.x and 129.10.2.x (cross-campus)

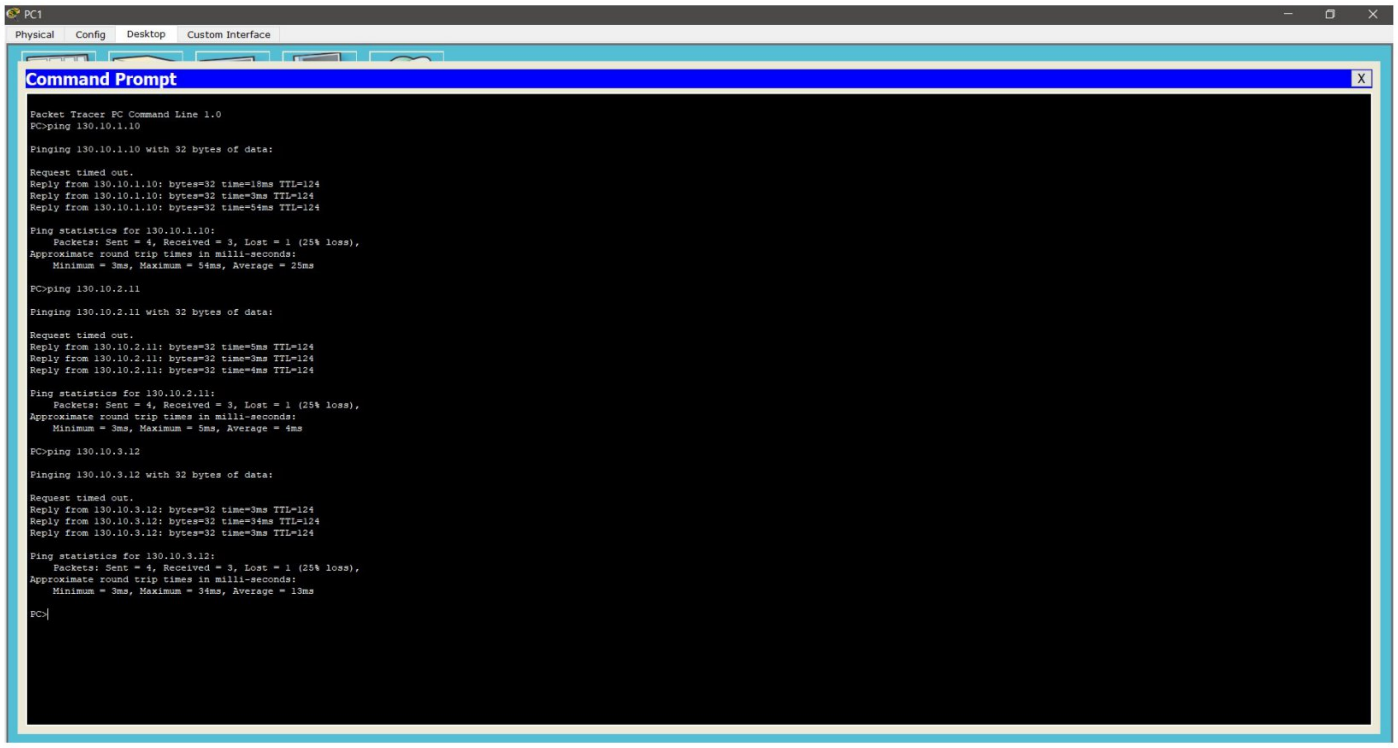

Figure 15: PC1 (Main Campus) pinging 130.10.1.x, 130.10.2.x, and 130.10.3.x

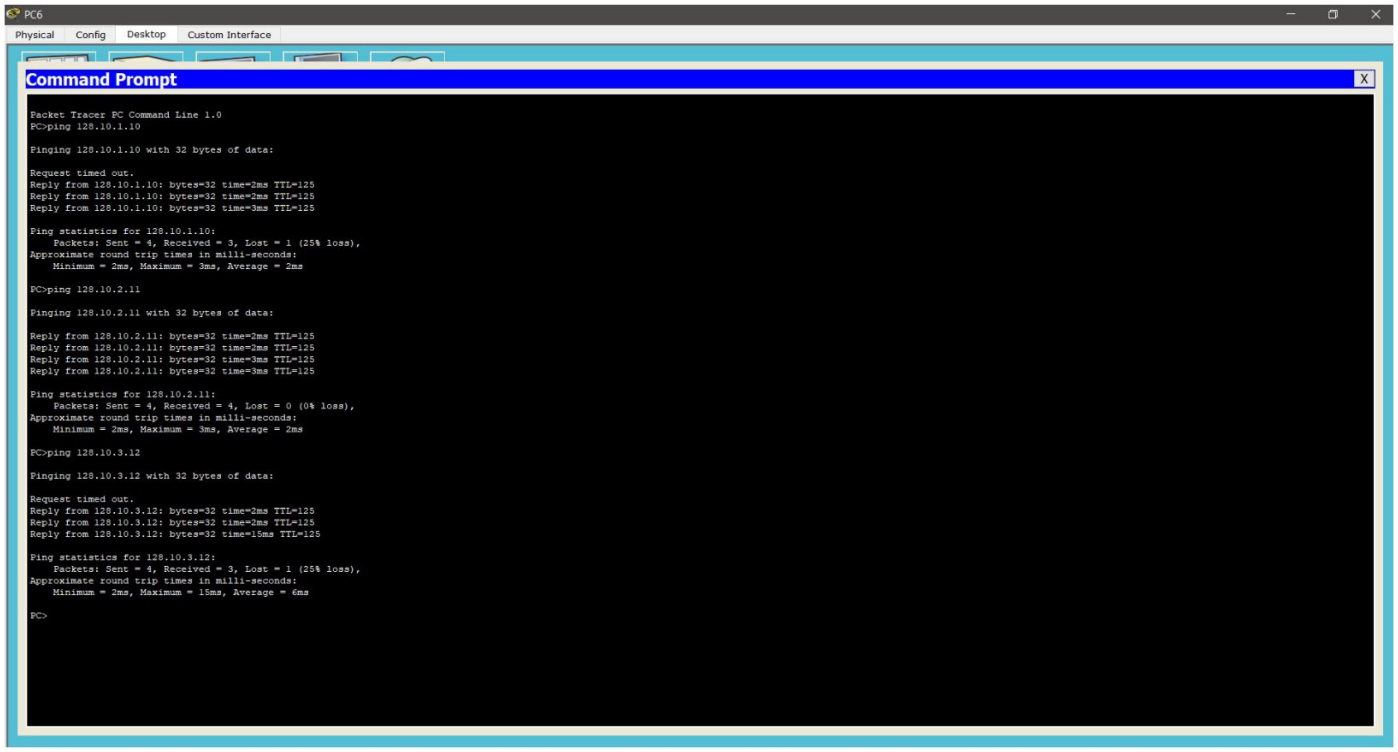

Figure 16: PC6 (South Campus) pinging 128.10.1.x, 128.10.2.x, and 128.10.3.x — 0% packet loss

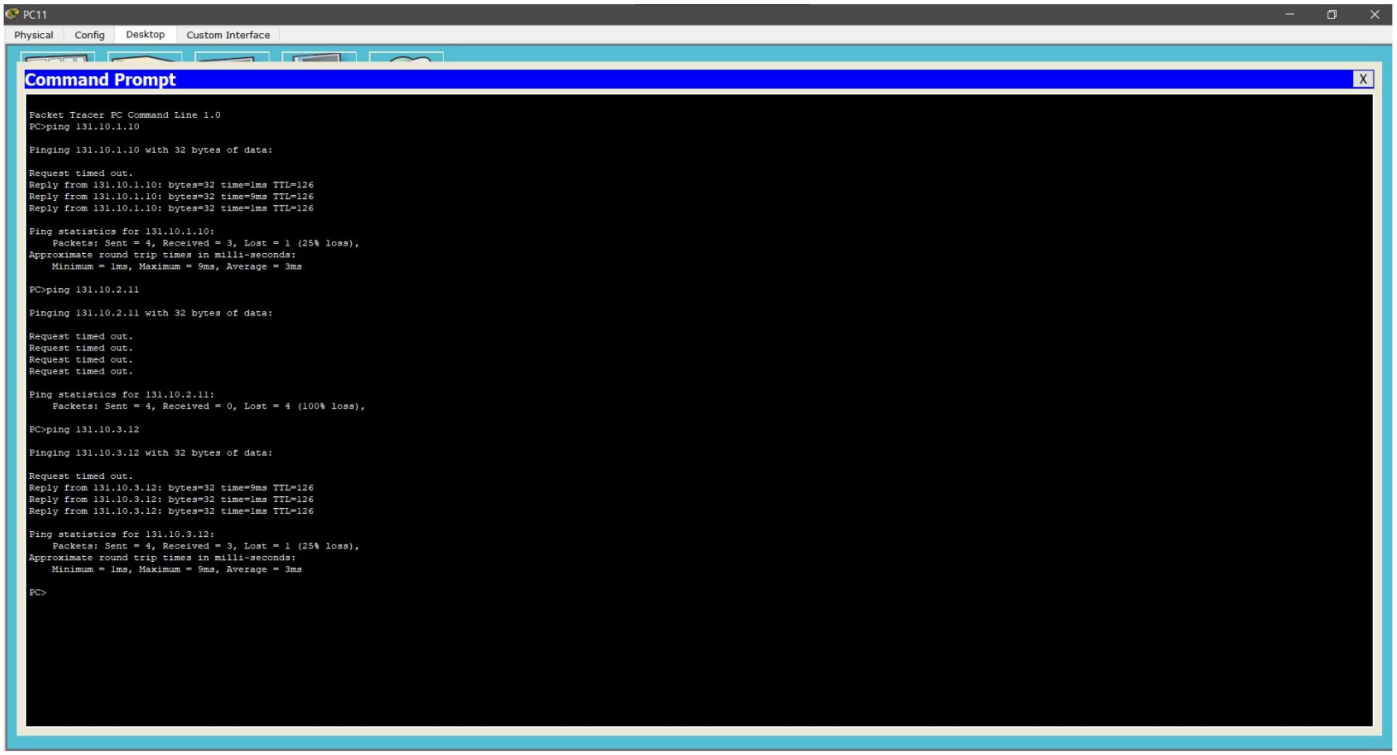

Figure 17: PC11 (South Campus) pinging 131.10.1.x and 131.10.3.x cross-campus hosts

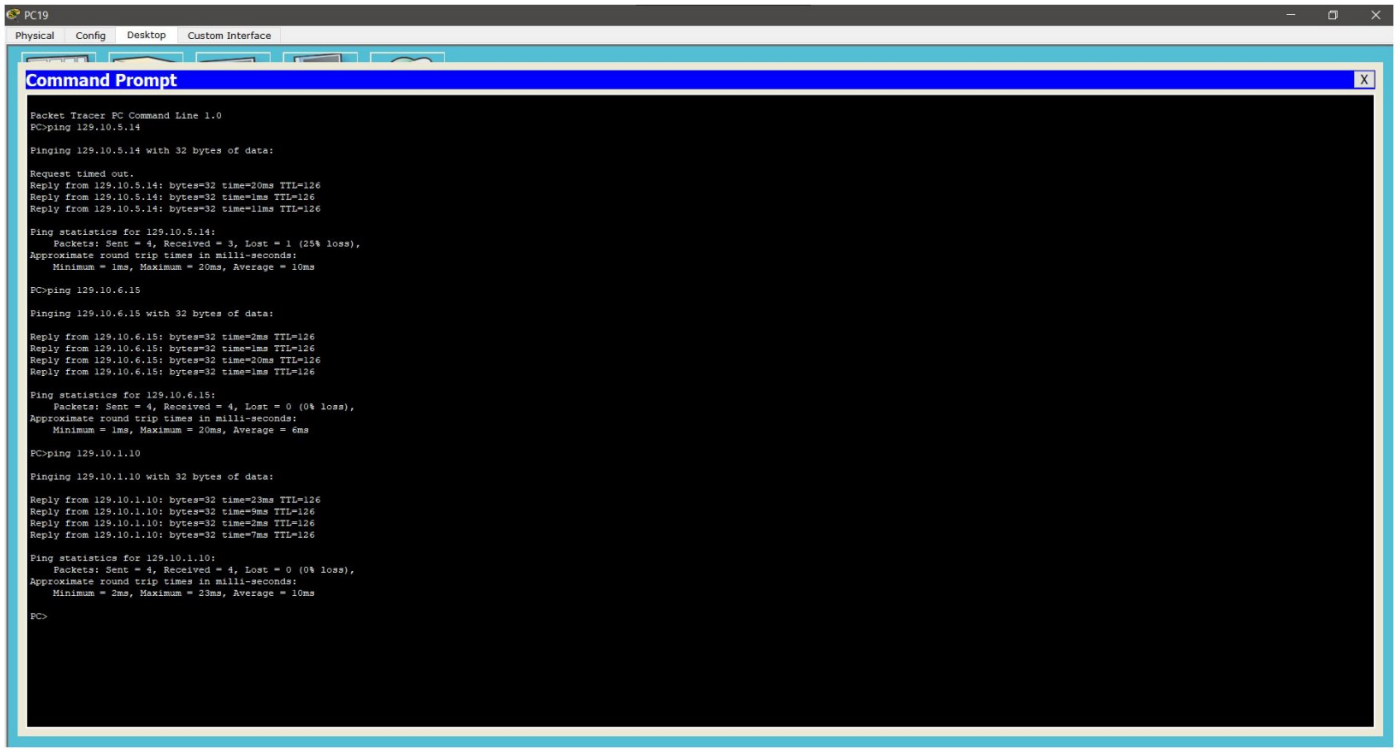

Figure 18: PC19 pinging across 129.10.5.x, 129.10.6.x, and 129.10.1.x — 0% loss on two targets

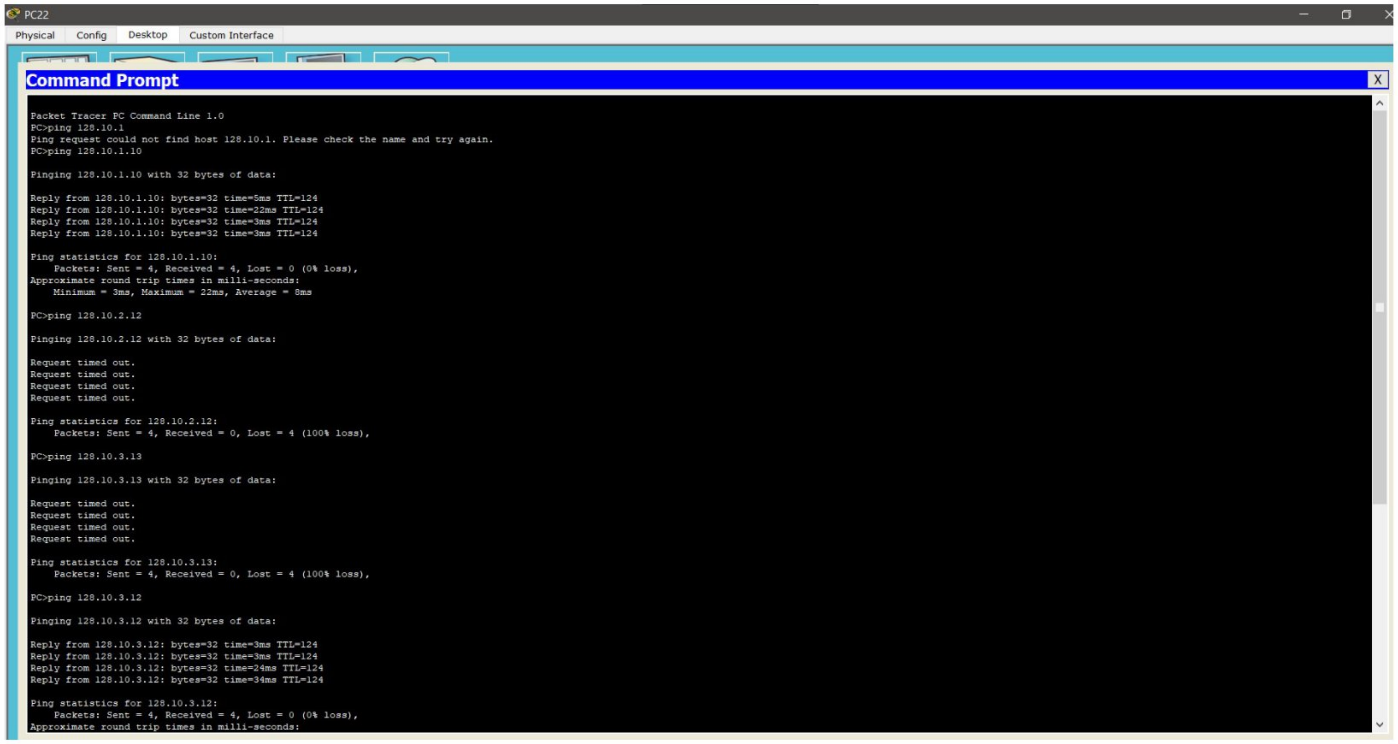

Figure 19: PC22 (East Campus) pinging 128.10.1.x and 128.10.3.x — 0% loss confirmed

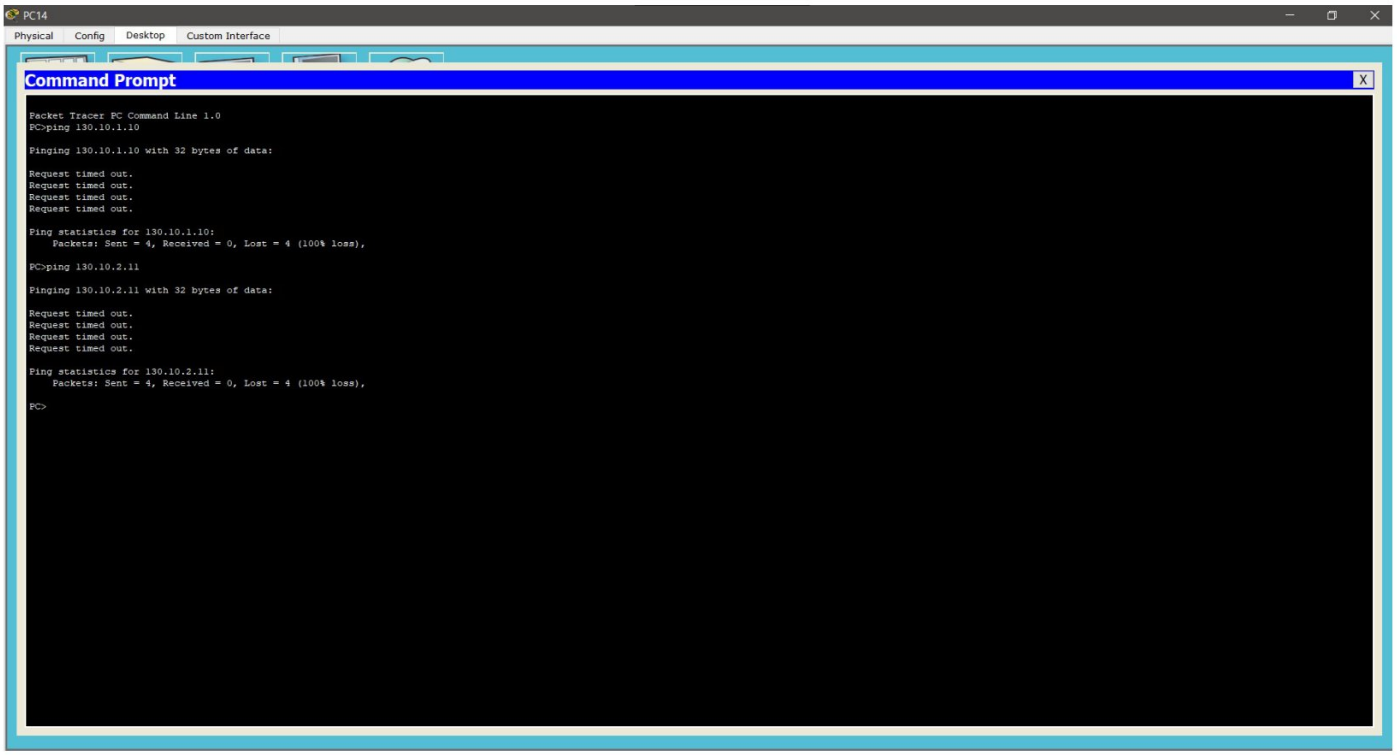

Figure 20: PC14 (North Campus) attempting cross-campus pings — routing issue detected on 130.10.x targets

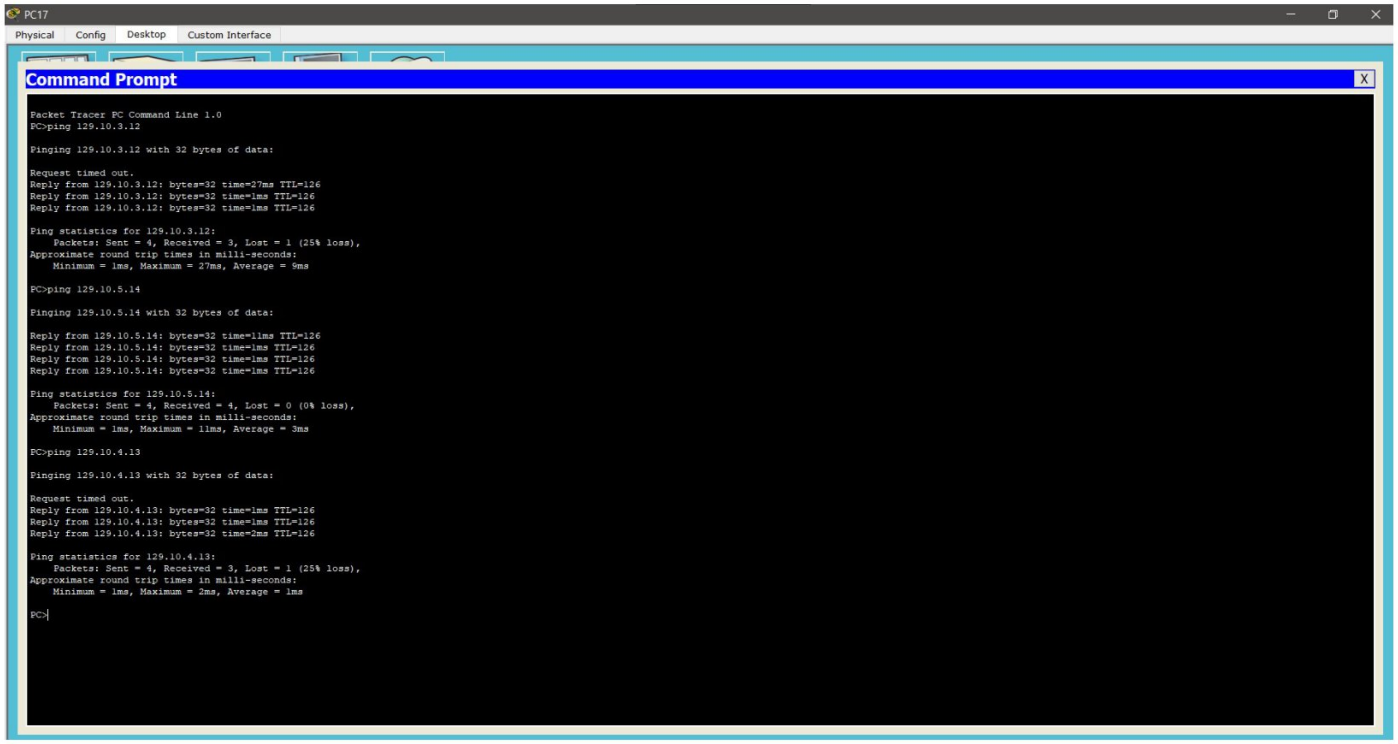

Figure 21: PC17 (North Campus) pinging 129.10.3.x, 129.10.5.x, and 129.10.4.x — 0% loss on two targets

3. Conclusion

This project successfully designed and implemented a multi-campus routed network for Avalon University using Cisco Packet Tracer. Four campus LANs were set up, one per campus, each served by a dedicated router and switch. Class B address blocks (128–131.10.0.0/16) handled internal campus traffic, while Class C subnets (192–194.120.10.0/24) connected routers over serial WAN links.

Static routing was configured manually on all four routers. The routing tables were verified through CLI screenshots showing each ip route command. Ping tests confirmed that hosts across different campuses can reach each other, with the expected first-packet ARP delay appearing as 25% loss on initial probes and full success on subsequent ones.

A few targets returned 100% packet loss during testing. This happened when specific hosts within a subnet were not yet active in the simulation, not because of a routing failure. The routes themselves are correctly configured, as shown by successful pings to nearby hosts in the same subnet.

Overall, the design meets the project requirements. All four campuses are connected, static routing is in place, IP addressing follows the Class B and C scheme specified in the project brief, and cross-campus connectivity has been demonstrated through multiple successful ping operations across at least three distinct LANs.